PROJECT CONCEPT & SYSTEM DESCRIPTION

JanSewa AI Nepal

Integrated Digital Governance & Public Service Intelligence Platform

A proposed AI/ML and Data Science-enabled platform for citizen services, ward offices, municipalities, district administration and related public-sector workflows.

Prepared as a concept document for product planning, prototype development, stakeholder discussion and potential pilot exploration.
This document does not imply government approval, endorsement or integration.

Contents

This document describes the project concept, modules, workflows, AI components, architecture, safeguards and implementation approach.

Section	What it covers
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2. Offices and portals	Ward, municipality, DAO/CDO and coordination portals
3. Citizen portal	Registration, applications, tracking, complaints and notices
4. Staff and officer portal	Work queues, document review, file movement and permissions
5. AI/ML modules	Document intelligence, chatbot, complaint classification and forecasting
6. Data Science dashboards	Service demand, processing time, workload and project analytics
7. Technical architecture	Frontend, backend, database, AI services and storage
8. Database and workflows	Core entities and example application lifecycle
9. Security and privacy	Access control, audit, data minimization and reliability
10. Roadmap and budget	Phased implementation and indicative cost bands
11. Government pilot proposal	Evaluation metrics, risks and next steps

Working name: JanSewa AI Nepal. The name is a provisional brand concept and should be checked for domain, company-name and trademark availability before commercial use.

1. Executive Summary

JanSewa AI Nepal is a proposed digital governance platform intended to make selected public services easier to access, track and manage. It connects citizen-facing services with authorized office workflows and provides reporting tools to help public officials understand application volumes, backlogs, complaints and service-delivery trends.

The platform is designed for a modular rollout. A first prototype can focus on one ward office and its municipality, then add district-level workflows after stakeholder requirements, legal authority, data-sharing rules and operating processes have been confirmed.

Core objectives

- Provide a citizen portal for service information, applications, document uploads, status tracking and complaints.
- Digitize selected office workflows, assignments, reviews, corrections, referrals and authorized decisions.
- Reduce repetitive data entry and make application progress more visible.
- Use AI to assist with document extraction, service FAQs and complaint categorization, with human review.
- Use Data Science to analyze service demand, processing times, pending work and local development records.
- Offer secure, role-specific portals rather than giving every office or employee unrestricted access to all records.

Expected value

- Citizens can check application status without repeatedly visiting an office solely to ask for updates.
- Staff can see assigned tasks, missing information and overdue items in one work queue.
- Managers can review aggregate service statistics and identify operational bottlenecks.
- A common technical foundation can support additional service modules over time.

Important scope note	This is a proposed concept, not an existing government service. Official deployment, integrations, legal validity of electronic documents, data exchange and procurement require approval from the relevant authorities.

2. Offices and Portals

Each office type should have a separate portal and configurable workflow. The software must reflect the legal authority and responsibilities of the actual office; it must not imply that one office can exercise another office's powers.

Portal	Potential functions
Ward Office	Recommendation requests; residence-related requests; local complaints; document checklists; application review; local development records; ward-level service reports.
Municipality / Rural Municipality	Ward-wise service monitoring; department task assignment; local revenue and receipt records; sanitation and waste complaints; infrastructure project tracking; budget and expenditure summaries.
District Administration Office / CDO workflows	Authorized public grievances; official correspondence and file tracking; referrals; follow-up reminders; meeting and task management; district-level summary reports.
District coordination	Inter-office referrals and acknowledgements; assigned tasks and deadlines; approved reports; aggregate district service trends.

Possible future modules

- Public notice and announcement publishing.
- Appointment scheduling for services that support appointments.
- Local development project milestone tracking.
- Information-request workflows where legally applicable.
- Configurable forms and service catalogues maintained by authorized administrators.

Service lists must be validated with each participating office. Some registrations, certificates, permits, fees and recommendations may follow specific national or local systems and legal procedures.

3. Citizen Portal

The citizen portal should be mobile-first, clear and available in Nepali and English. Romanized Nepali support may be added where technically feasible. The interface should guide people through each service without requiring them to understand internal government organization charts.

3.1 Registration and profile

- Account registration using an approved authentication method, such as verified mobile or email where appropriate.
- Basic profile and address information, with municipality, ward and district selection where required.
- A personal dashboard for submitted applications, requests and notifications.
- Privacy controls and consent notices where applicable.

3.2 Online applications

- Service catalogue with eligibility notes, required documents, applicable fees and process guidance.
- Configurable application forms and save-as-draft capability.
- Secure uploads for PDF, JPG and PNG, with file-size and type validation.
- Application acknowledgement and a unique tracking ID.
- Correction requests when information is incomplete.

3.3 Application tracking

Status	Meaning
Draft	Application started but not submitted.
Submitted	Application received by the system.
Under review	Assigned staff or authorized office is reviewing it.
Additional information required	Applicant needs to provide corrections or documents.
Referred	Sent to another authorized office or department.
Approved / Rejected	Decision recorded by an authorized official, with applicable reasons.
Completed	Workflow completed under the configured service process.

Illustrative tracking ID: JSN-WARD05-2026-000123. Production IDs should be generated safely and must not expose unnecessary personal information.

3.4 Complaints, suggestions and notifications

- Complaint category, description, optional photograph and location where suitable.
- Acknowledgement and status tracking, with routing to an authorized department.
- Email, SMS or in-app notifications through supported providers.
- Public notices and service announcements, with clear source and publication date.

4. Staff and Officer Portal

The staff portal organizes office work into queues, assignments and recorded decisions. It should be configurable so that each office can use only the workflows that match its responsibilities.

4.1 Work dashboard

- New applications and cases assigned to the logged-in user.
- Document checks, correction requests and referrals.
- Pending tasks, due dates and overdue reminders.
- Appointments and recent activity.
- Filters by service, date, status, ward and department according to permissions.

4.2 Application workflow

1. Staff opens the submitted application and checks the provided information.
2. System shows the required-document checklist and any AI-generated flags.
3. Staff verifies the original documents and requests corrections when needed.
4. Application is assigned or forwarded to the next authorized reviewer.
5. Competent official records the decision and any required notes.
6. Citizen receives an appropriate status update and, where applicable, the final document through the authorized process.

4.3 Digital file and correspondence management

- File number, subject, originating office and related application.
- Attachments, assignments, deadlines, notes and movement history.
- Search and archive subject to retention and access policies.
- Official signatures, seals and document validity handled according to applicable requirements.

4.4 Roles and permissions

Role	Typical access
Citizen	Own applications, documents, complaints and notifications.
Data entry operator	Enter and update records within assigned permissions.
Authorized ward staff	Process assigned ward workflows and review documents.
Department officer	Review, refer and act on records within the department.
Municipality administrator	Authorized municipal oversight and reports.
District officer	Authorized district workflows and reports.
Technical administrator	System configuration and account administration; sensitive data access remains separately controlled.

5. AI and Machine Learning Modules

AI features should be added to specific, measurable tasks. AI outputs are assistive suggestions and should be reviewable; they should not autonomously approve or reject government entitlements, registrations, recommendations or legal decisions.

5.1 AI Document Intelligence

- OCR-based extraction of text from supported Nepali and English documents.
- Extract candidate fields such as dates, names and reference numbers for staff review.
- Compare uploaded files against a configured checklist and flag possible missing items.
- Flag low-quality scans or unreadable text.
- Show extracted text alongside the original file so staff can verify it.

Methods: OCR, document classification and information extraction. Performance depends on scan quality, scripts, document formats and validation data.

5.2 Nepali Government Service Assistant

- Answer common service questions in Nepali and English; add Romanized Nepali support if validated.
- Explain required documents and application steps from an approved knowledge base.
- Provide source links and dates for service information.
- Direct users to the appropriate service form or office information.
- Decline to guess when the approved source does not contain an answer.

Methods: Large language model (LLM), retrieval-augmented generation (RAG), multilingual retrieval and source-grounded responses.

5.3 Smart Complaint Classification

- Suggest categories such as road, water supply, sanitation, infrastructure or other services.
- Extract potential location terms when appropriate.
- Route low-confidence cases to manual review.
- Keep classification explainable and avoid punitive decisions based on identity or inferred traits.

Methods: Text classification, NLP and confidence-based review.

5.4 Predictive Service Analytics

- Analyze historical application counts and processing times.
- Forecast service demand where sufficient quality historical data exists.
- Estimate workload trends to support planning and staffing.
- Display model uncertainty and distinguish forecasts from actual recorded figures.

Methods: pandas, scikit-learn, regression and time-series models. Forecast quality must be evaluated against a baseline before operational use.

5.5 Local Development Intelligence

- Map aggregated complaint patterns by ward or locality.
- Track project milestones, approved budgets and recorded expenditure.
- Compare service request volumes over time.

- Restrict sensitive citizen information from public maps and reports.

6. Data Science and Executive Analytics

Analytics should help officials understand workloads and trends without misrepresenting what the data can prove. Reports should identify the period, source, completeness and any limitations of the data.

Dashboard / Report	Purpose
Ward Performance	Compare service volumes, backlog and processing-time trends by ward.
Pending Application Report	Identify pending and overdue cases for operational follow-up.
Service Demand	Understand which services receive more applications over time.
Complaint Analysis	Summarize categories, locations and response patterns.
Project Progress	Track milestones, dates, approved budget and recorded expenditure.
Executive Summary	Provide aggregate views for authorized decision-makers.

Recommended KPI definitions

- Applications received and completed within a clearly specified reporting period.
- Pending applications at the end of the period.
- Median and average elapsed processing time, with exclusions documented.
- Correction-request rate and reasons, where reliably recorded.
- Complaint acknowledgement and resolution times.
- Workload by office, department, service and ward.
- Data completeness, model performance and system availability.

Illustrative dashboard data

A prototype may use clearly labelled synthetic records to demonstrate charts and filters. Synthetic figures must never be presented as official Nepal government statistics or as evidence of real-world impact.

Month	Sample requests
Jan	120
Feb	155
Mar	138
Apr	190
May	174
Jun	220

7. Technical Architecture

A modular monolith is a reasonable starting point for a prototype: one backend application divided into clear modules, with separate AI services where useful. This avoids premature complexity while keeping the design extensible.

Layer	Suggested components	Responsibilities
Frontend	React, Vite, Tailwind CSS	Citizen, staff, ward, municipality and district portals.
API and access	Python FastAPI	Validation, authentication, permissions, workflow APIs and audit events.
Core database	PostgreSQL	Users, organizations, services, applications, statuses and records.
AI/ML services	OCR, NLP, RAG, scikit-learn	Document extraction, service FAQ assistant, complaint categorization and forecasting.
Analytics	pandas, NumPy, Recharts or Plotly	Aggregation, reporting, charts and model evaluation.
Files and storage	Encrypted object storage	Access-controlled documents, backups and retention policies.
Maps	Leaflet and suitable geographic datasets	Authorized geographic summaries and project locations.

High-level data flow

Step	Flow
1. User interface	Citizen or staff submits a permitted request through the portal.
2. Secure API	Authentication, authorization, validation and audit event recording.
3. Core services	Application workflow and records stored in PostgreSQL.
4. Optional AI service	Document or text is processed for assistive output under approved data rules.
5. Human review	Authorized staff checks AI flags and records official actions.
6. Reporting	Aggregated data updates dashboards and reports.

8. Database Design and Example Workflow

The data model should support multiple organizations and keep each record linked to the responsible office. Sensitive fields and files should be protected separately from aggregate analytics datasets.

Entity	Key information
Users	Login identity, account state and authentication references.
Citizen Profiles	Minimum necessary personal details and preferences.
Organizations	Ward, municipality, district office and department.
Roles & Permissions	Role assignments, office scope and allowed actions.
Services	Service type, requirements, instructions and workflow configuration.
Applications	Tracking ID, applicant reference, service, status and timestamps.
Application Documents	Secure file reference, document type and review status.
Workflow History	Status changes, assignments, decisions and timestamps.
Complaints	Category, description, suitable location and status.
Official Tasks	Assigned role or user, due date and completion information.
Development Projects	Project details, approved budget, milestones and progress.
Notifications	Recipient, channel, message type and delivery state.
Audit Logs	Actor, action, record reference, timestamp and security context.
Analytics Data	Aggregated metrics and approved historical trends.

Example: recommendation application

1. Citizen selects a service and reads the current approved checklist.
2. Citizen submits the form and uploads required documents.
3. System creates a tracking ID and acknowledgement.
4. AI may extract fields or flag possible missing information for review.
5. Authorized staff verifies the documents and requests corrections if needed.
6. Application moves to the competent official according to the office workflow.
7. Authorized official records the decision and applicable reason.
8. Citizen sees the updated status and receives an appropriate notification.
9. Aggregate metrics update without exposing private citizen information.

9. Security, Privacy and Reliability

Because the platform may handle identity documents, administrative requests and complaints, security and privacy must be designed from the beginning, not added at the end.

- Role-based, organization-scoped access; users should see only records they are authorized to access.
- Encryption in transit and at rest, secure password storage and appropriate multi-factor authentication for privileged users.
- Audit trails for sensitive views, downloads, edits, referrals and decisions.
- Secure file upload scanning, file-type validation, size limits and controlled download links.
- Rate limiting, input validation, secure session handling and protection against common web vulnerabilities.
- Backups, recovery tests, monitoring and an incident response procedure.
- Data retention, correction and deletion processes consistent with applicable law and official records requirements.
- Privacy review before sending documents to external AI services; avoid training models on sensitive records without explicit authorization.
- Data minimization and de-identification for analytics wherever appropriate.
- Accessibility, mobile usability and low-bandwidth considerations for citizens.

AI governance safeguards

- AI outputs must be labelled as suggestions, not official decisions.
- Users should be able to review the source document and correct extracted fields.
- Chatbot responses should use an approved, maintained knowledge base and show sources where possible.
- Low-confidence or high-impact cases should go to a human reviewer.
- Models should be tested for accuracy across relevant document types and language variants.
- Monitor errors, maintain model versions and provide a fallback to manual processing.

Deployment caveat

A production government deployment should undergo legal, security, privacy, accessibility and operational reviews. Hosting, data residency, government identity systems, electronic signatures, official integrations and procurement must follow the relevant authority's requirements.

10. Development Roadmap

Phase	Indicative timing	Deliverables
1. Discovery and requirements	Weeks 1-2	Stakeholder interviews, service workflows, roles, prototype scope and data model.
2. Core application	Weeks 3-5	Citizen portal, application submission, uploads, tracking IDs and staff queues.
3. Office workflow and analytics	Weeks 6-7	Ward/municipality workflows, complaints, tasks, reporting and dashboards.
4. AI/ML modules	Weeks 8-9	Document extraction, source-grounded FAQ assistant, complaint classification and basic analysis.
5. Testing and demonstration	Weeks 10-12	Security and access tests, synthetic scenarios, mobile testing, model evaluation and demo presentation.

The 12-week schedule is an illustrative estimate for a focused prototype with a small, capable team. Solo development, complex integrations, stakeholder approvals and production-grade security can increase the timeline.

Recommended first release

- One ward office and one municipality, using synthetic data.
- One complete application workflow from submission to authorized decision.
- Complaint tracking and basic staff task management.
- One AI document-checklist assistant with human verification.
- One analytics dashboard with clearly labelled demo metrics.
- Separate roles and test accounts for citizen, staff and administrator.

11. Indicative Budget and Operating Costs

The following ranges are rough planning estimates in Nepalese rupees (NPR), not vendor quotations or official government rates. Actual costs depend on scope, team, hosting, AI usage, security requirements, integrations, support and procurement.

Version	Indicative range	Scope
Prototype	NPR 30,000-100,000	Demo with synthetic data, selected screens, basic workflows and limited AI proof-of-concept.
Functional MVP	NPR 150,000-500,000	Working application, database, role-based portals, selected AI features and reporting.
Limited production pilot	NPR 800,000-2,500,000+	A limited-office pilot with stronger security, testing, deployment, training and support.
Multi-office deployment	Estimate after discovery	Requires detailed requirements, integration planning, infrastructure and procurement analysis.

Budget categories to calculate

- Product design, frontend and backend development.
- AI/ML development, evaluation and any model/API usage.
- Hosting, database, storage, backups and monitoring.
- Security testing, privacy review and legal/procurement support.
- Data preparation, migration and approved integrations.
- Staff training, documentation, user support and maintenance.

Costs can be lower in cash terms if the founder develops the prototype directly, but development time, testing and ongoing maintenance remain real costs.

12. Government Pilot Proposal and Evaluation

A funding or pilot proposal should focus on a defined public-service problem, a limited scope, measurable outcomes and a realistic operating plan. The demo alone does not imply government endorsement or a procurement commitment.

Recommended proposal sections

- Problem statement supported by stakeholder interviews or documented workflow observations.
- Proposed solution and specific services included in the pilot.
- Beneficiaries: citizens, staff and authorized decision-makers.
- Working prototype demonstration and sample workflow.
- Technical architecture, AI methods and limitations.
- Implementation timeline, responsibilities and budget breakdown.
- Security, privacy, data-sharing and records-management plan.
- Training, maintenance, support and sustainability plan.
- Pilot evaluation approach, baseline and success criteria.

Suggested evaluation indicators

Indicator	How to evaluate
Application tracking coverage	Share of pilot applications with a usable status history.
Processing time	Compare baseline and pilot period using a consistent definition.
Pending workload	Track open and overdue cases by service and office.
Document assistant quality	Measure field-extraction accuracy and checklist flag precision on approved test data.
Complaint classification	Measure precision, recall and manual correction rate.
Chatbot reliability	Sample answers against an approved reference set; track unsupported answers.
Usability	Collect feedback from citizens and staff using structured tasks.
Security and availability	Track test findings, incidents and service uptime.

All prototype results using synthetic data should be labelled as simulated. Real-world improvement claims require measured pilot data and a suitable comparison baseline.

13. Key Risks and Mitigations

Risk	Mitigation
Unclear office responsibilities	Validate each workflow with the responsible office; configure permissions and approval steps accordingly.
Poor or incomplete source data	Start with a limited scope, document data quality and avoid unreliable forecasts.
AI mistakes or hallucinations	Use source-grounded answers, confidence thresholds, human review and model evaluation.
Sensitive personal information exposure	Minimize data, restrict access, encrypt records and audit sensitive actions.
Connectivity or digital literacy barriers	Use mobile-first design, simple language, low-bandwidth pages and retain assisted/offline alternatives as needed.
Integration and procurement delays	Keep the prototype independently demonstrable and plan integrations only after authorization.
Ongoing maintenance costs	Budget for support, backups, training, upgrades and responsible ownership from the beginning.

Immediate next steps

1. Choose one pilot service and confirm its actual workflow with a relevant office.
2. Create wireframes for citizen, staff and executive dashboards.
3. Define roles, permissions, data fields and a simple application state machine.
4. Build the core workflow with synthetic data before adding AI.
5. Add one AI feature and evaluate it against a small, reviewed test set.
6. Prepare a short demo video, technical summary, budget and pilot evaluation plan.

Project vision	Build a practical, secure and measurable digital service platform that helps citizens navigate public services and helps authorized offices manage work more transparently. Start small, validate with real stakeholders and expand only when the workflow, data and governance requirements are clear.